

by Michael Penn cited

$$f^{-1}(x) = \frac{1}{f(x)}$$

Inverse Equals Inverse Function

Cheng, Dasgupta, Ebanks
Kinch, Larson, McFadden

When is

$$f^{-1}(x) = \frac{1}{f(x)}?$$

when f is real valued.

$$x = f(f^{-1}(x)) = f\left(\frac{1}{f(x)}\right)$$

$$x = f^{-1}(f(x)) = \frac{1}{f(f(x))}$$

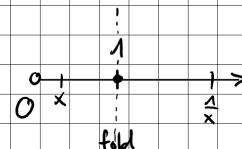
$$\Rightarrow \boxed{f(f(x)) = \frac{1}{x}}$$

$$\Rightarrow f \circ f \circ f \circ f(x) = x$$

Goal: Find $f: (0, \infty) \rightarrow (0, \infty)$, $f(f(x)) = \frac{1}{x}$.

Define: $\sim$ on $(0, \infty)$ by $x \sim y$ if $\frac{x}{y} = \frac{y}{x}$

Note: $[x] = \{x, \frac{1}{x}\}$, $[2] = \{2, \frac{1}{2}\}$, $[4] = \{4, \frac{1}{4}\}$,
 $[1] = \{1\}$



Partition: $(0, \infty) / \sim = U \cup V \cup \{1\}$ with $|U| = |V|$

ex: $U = \{[\frac{1}{2}], [3], \dots\}$, $V = \{[\frac{1}{3}], [\pi], \dots\}$

take any bijection: $g: U \rightarrow V$

$$\varphi: U \cup V \rightarrow U \cup V \quad \varphi([x]) = \begin{cases} g([x]) & [x] \in U \\ g^{-1}([x]) & [x] \in V \end{cases}$$

so far $[x] = \{x, \frac{1}{x}\}$, $\varphi: U \cup V \rightarrow U \cup V$ (from g)

Axiom of Choice $\rightarrow$ Choice-Function C

$$C: U \cup V \rightarrow (0, \infty) \setminus \{1\}, C([x]) = x \text{ or } \frac{1}{x}$$

Define:

$$f(x) = \begin{cases} C(\varphi([x])) & \text{if } [x] \in U \quad C([x]) = x \\ C([x]) & \text{if } [x] \in V \quad C([x]) = \frac{1}{x} \\ 1 & x = 1 \\ \frac{1}{C(\varphi([x]))} & \text{if } [x] \in U \quad C([x]) = \frac{1}{x} \\ C(\varphi([x])) & \text{if } [x] \in V \quad C([x]) = x \end{cases}$$

satisfies $f(f(x)) = x$

example:

$$f(x) = \begin{cases} \frac{x}{1-x} & x \in [\frac{1}{2n+1}, \frac{1}{2n}) \\ \frac{x+1}{x} & x \in [\frac{1}{2n}, \frac{1}{2n-1}) \\ 1 & x = 1 \\ x-1 & x \in (2n, 2n+1] \\ \frac{1}{x+1} & x \in (2n-1, 2n] \end{cases} \quad n \in \mathbb{N}$$

case: let $x \in (2n-1, 2n] \Rightarrow \frac{1}{x+1} \in [\frac{1}{2n+1}, \frac{1}{2n})$

$$\begin{aligned} \Rightarrow f(f(x)) &= f\left(\frac{1}{x+1}\right) = \frac{\frac{1}{x+1}}{1 - \frac{1}{x+1}} \cdot \frac{(x+1)}{(x+1)} \\ &= \frac{1}{x+1-1} = \frac{1}{x}, \text{ hence } f(f(x)) = \frac{1}{x} \end{aligned}$$

An "intuitive" take:

$$(f \circ f \circ f \circ f)(x) = x \quad \text{but } f \circ f(x) \neq x$$

note: $g(x) = \frac{1}{x} \quad g \circ g(x) = x \quad (\mathbb{R} \setminus \{0\})$

$$h(x) = -x \quad h \circ h(x) = -(-x) = x \quad (\mathbb{R})$$

$$\hookrightarrow f(x) = \begin{cases} -x & x \in (0, \infty) \\ -\frac{1}{x} & x \in (-\infty, 0) \end{cases}$$

Homework • What if f is complex valued?

• Similar: $f \circ f(x) = [f(x)]^2$